

The visible side

Mathematics before the session

There are three two-sided cards: RR has faces 1 and 2; RB has faces 3 and 4; BB has faces 5 and 6. Their face colors in order are R, R, R, B, B, B. A ticket selects one specific face to show. Under the six-ticket rule every ticket has probability $1/6$, with replacement and fresh mixing between rounds.

Ticket	Showing color	Hidden color
1 or 2	R	R
3	R	B
4	B	R
5 or 6	B	B

If the only clue is red showing, the surviving equally likely tickets are 1, 2, 3. Two hide red and one hides blue, so $P(\text{hidden red} \mid \text{showing red}) = 2/3$. Likewise, given blue showing, tickets 4, 5, 6 remain and hidden blue has probability $2/3$. The two possible card types after a red clue are not equally likely: RR supplied two eligible faces.

Changing the chooser changes the answer. With only tickets 1 and 3 selected uniformly, red always shows; one ticket hides red and one blue. Each hidden color then has probability $1/2$. The visible clue is the same, but its production mechanism is different. This is the conditional-sample-space point of Bertrand's box paradox.

Exact design condition

For a subset S of the six distinct tickets chosen uniformly, a red clue is fair precisely when ticket 3 is present and exactly one of tickets 1 and 2 is present. Any subset of $\{4,5,6\}$ may also be present, since those tickets cannot give a red clue. Thus there are $2 \times 2^3 = 16$ fair subsets; only two are two-ticket cups. Keeping only whole cards can never achieve this condition, because faces 1 and 2 must then be kept or removed together. A cup that never shows red is excluded.

Assumptions and discoveries by band

The face number, card identity, unshown side, orientation process, and distinctive physical marks must stay hidden from the guesser. Revealing a number determines the hidden color exactly. Equal-size tickets and cards, a screen, and identical covers make "color alone" an executable information rule. Repeated outcomes do not prove a conditional probability.

K-1: sort face possibilities, remove or add tickets, and compare a color clue's surviving cases.

Grades 2-3: count all six face outcomes and contrast the two chooser rules. **Grades 4-5:** diagnose unequal conditional weights, information leaks, and characterize all fair subsets.

Running the session

Materials and preparation

Make three same-size, opaque, two-sided cards per table: RR numbered 1/2, RB numbered 3/4, BB numbered 5/6. Number each face in the same corner. Prepare six same-size tickets numbered 1-6, a cup, a folded folder as screen, and identical opaque covers for face marks. For ten children, three adult-led table kits suffice; five kits allow paired play. Use R/B letters along with colors and check that no reverse-side printing shows through.

Print four pages per band. Readiness is matching two-sided objects, sorting by a clue, and small counts; fractions are optional. Adults may handle hidden orientation while children arrange visible ticket models for the proof. Give everyone time to inspect the known cards before hiding anything. Prepared, unpiloted material.

Short shared launch

Show that ticket 4 selects the blue face of RB and its other face is red. Then demonstrate behind the screen: draw the ticket, orient its named face upward, hide the ticket, cover the number identically, and reveal only color. Return and mix the ticket after each round. Ask which tickets could have caused the displayed color, rather than asking children to guess the teacher's intentions.

Executing the added ticket cases

In K-1 Problem 4, an added ticket shows its own numbered face, even if it is blue. Compare hidden colors only when red is showing; set blue-showing rounds aside. The two-ticket chooser always shows red, but an expanded cup may not. Use the complete ticket set to explain why blue-showing additions cannot change a red-clue comparison.

A flexible hour and optional hints

0-8 minutes: inspect the cards and demonstrate the screen. **8-23:** sort all six tickets and investigate a red clue. **23-33:** compare a blue clue or ticket removals. **33-36:** movement break with three children representing paired faces. **36-49:** compare the two-ticket chooser or design a cup. **49-57:** show the surviving cases for a claim. **57-60:** reset kits. A complete three-survivor argument is a good endpoint; the 16-subset description can remain with the oldest group.

Only if needed: "Could that ticket show red?" "Does RR have one red face or two?" "Which added tickets can survive a red clue?" "What information does that number reveal?" Stop trials after a few rounds and use the complete ticket set; no long frequency ledger is needed.

Source and adaptation

Background: Brian Mintz, Dartmouth Graduate Student Seminar, Fall 2022, *Probability Paradoxes: Eleven Enlightening Examples*; seminar slides, with provenance on the Dartmouth seminar page. The six-ticket protocol, subset designs, and exact answers here are independently derived classroom adaptations.

K to 1 answer key

Problem 1

Hidden red: tickets **1, 2, 4**. Hidden blue: **3, 5, 6**. Same showing color but different hidden colors: 1 and 3, or 2 and 3, both show red; 4 and 5, or 4 and 6, both show blue. These four unordered pairs are all such choices.

Problem 2

Given red, hidden red has more ways: tickets 1 and 2 versus ticket 3. Removing **1 or 2** makes a tie, one surviving red-clue ticket for each hidden color. Removing 3 leaves only hidden red; removing 4, 5, or 6 changes no red-clue possibility.

Problem 3

Restore all tickets first. Given blue, hidden blue has two ways (5,6), hidden red one (4). Remove **5 or 6** to make a tie. Removing 4 leaves only hidden blue; removing 1, 2, or 3 changes no blue-clue possibility.

Problem 4

With {1,3}, neither hidden color has more ways: ticket 1 hides red and ticket 3 hides blue. **Given red showing**, adding **2** favors hidden red, two ways versus one. Adding **4, 5, or 6** leaves a tie because the added ticket shows blue and cannot have produced the red clue.

If a child instead counts hidden colors over *all* rounds, adding 2 or 4 favors red 2-to-1; adding 5 or 6 favors blue 2-to-1. Those counts are correct for that different question. Name the conditioning clue before comparing answers.

Problem 5

The complete three-ticket choices are **{3,4,5}, {3,4,6}, {3,5,6}**. Ticket 3 must be included: it is the only red-showing face that hides blue. Tickets 1 and 2 must be excluded because they would show red and hide red. Choose the other two tickets from 4, 5, 6. Every listed cup sometimes shows red and then always hides blue.

Problem 6

Remove the **RB card**, meaning tickets 3 and 4, to make red showing always hide red. Remove the **RR card**, tickets 1 and 2, to make red showing always hide blue. In both cases a red clue remains possible. Removing BB leaves the original 2-to-1 red-clue weighting.

Problem 7

Exactly **{1,3}** and **{2,3}**. A fair red clue needs one red-showing ticket hiding red (1 or 2) and one hiding blue (3). A blue-showing ticket uses up one of the two places without contributing to either red-clue category, so no other two-ticket cup works.

Grades 2 to 3 answer key

Problem 1

Ticket	Showing	Hidden	Same color both sides
1	R	R	Yes
2	R	R	Yes
3	R	B	No
4	B	R	No
5	B	B	Yes
6	B	B	Yes

Thus tickets 1, 2, 5, 6 show a color matching the hidden side. The RR and BB cards each contribute two separate face choices.

Problem 2

Red is the better hidden-color guess. Red-showing tickets 1, 2, 3 are the surviving equally likely choices. Tickets 1 and 2 hide red; ticket 3 hides blue. Hence red has probability $2/3$ and blue $1/3$, given that red shows. Blue-showing tickets do not count in this conditional sample space.

Problem 3

The better guess changes to blue. Blue-showing tickets 4, 5, 6 survive: 4 hides red; 5 and 6 hide blue. Hidden blue has probability $2/3$. The rule “guess the showing color” wins two of three clue-compatible face outcomes for either clue.

Problem 4

They do not give the same hidden-color chances. Under the $\{1,3\}$ chooser, red always shows and one of two equally likely tickets hides each color, so hidden red has chance $1/2$. Under the six-ticket chooser restricted to red-showing rounds, tickets 1, 2, 3 remain, so hidden red has chance $2/3$. The words “red is showing” are the same in both experiments; the rule that generated those words differs.

Problem 5

Remove **RB** (tickets 3,4): red can still show and then certainly hides red. Remove **RR** (1,2): red can still show and then certainly hides blue. Remove **BB** (5,6): the red-clue odds stay 2-to-1 for red. None of the three whole-card removals gives equal hidden-color chances under a red clue.

Problem 6

The only pairs are **{1,3}** and **{2,3}**. Ticket 3 is the sole red-showing/blue-hidden outcome. To balance it, choose exactly one of 1 or 2. If the pair contains a blue-showing ticket instead, only one red-showing ticket remains and its hidden color is certain; if neither shows red, the required clue is impossible.

Grades 4 to 5 answer key

Problem 1

Given red showing, tickets 1,2,3 remain equally likely: hidden red $2/3$, hidden blue $1/3$. Given blue showing, tickets 4,5,6 remain: hidden red $1/3$, hidden blue $2/3$. Count face tickets rather than merely the two card types compatible with the clue.

Problem 2

The argument is unsupported. Before conditioning, each card has two of the six face tickets. But both RR tickets show red, whereas only one RB ticket does. After retaining red-showing rounds, RR therefore has twice RB's weight. The three surviving face outcomes, not the two surviving card names, are equiprobable.

Problem 3

The mark determines the hidden color: 1,2,4 hide red; 3,5,6 hide blue. Each number identifies its mate. Cover the mark, conceal the ticket, and hide the orientation process so the guesser receives only the displayed color.

Problem 4

Yes. With tickets 1 and 3, red always shows and the hidden-red chance is $1/2$. With all six tickets conditioned on red, it is $2/3$. One protocol gives each red-capable card one eligible face ticket; the other gives the RR card two. Both protocols are fully specified, so no assumption about a host's unstated preferences is needed.

Problem 5

No. Six hidden reds can occur under both rules, with conditional probabilities $(2/3)^6$ and $(1/2)^6$ respectively. Every six-color binary sequence is possible under both. The exact mechanisms prove the distributions differ; a short sample does not logically separate them.

Problem 6

Every solution has the form $\{1,3\} \cup T$ or $\{2,3\} \cup T$, where T is any subset of $\{4,5,6\}$. Explicitly T may be $\emptyset$, $\{4\}$, $\{5\}$, $\{6\}$, $\{4,5\}$, $\{4,6\}$, $\{5,6\}$, or $\{4,5,6\}$. These two families give all 16 subsets, without repeats.

Proof of completeness: among red-showing tickets, hidden blue has only ticket 3 available, so it must be included. For equal positive weight, exactly one hidden-red ticket, 1 or 2, must be included. Blue-showing tickets are unrestricted because conditioning on red removes them.

Problem 7

No. Keeping RR supplies both hidden-red red-showing tickets or neither; keeping RB supplies the one hidden-blue red-showing ticket or none. A positive tie would require exactly one of 1,2 plus 3, which whole-card selection cannot do. BB supplies no red faces and cannot repair the imbalance. Exclude a BB-only selection because red never shows.